

Problem Statement

3+ Weeks

Average time MSEs spend
identifying applicable standards

564+

BIS standards in Building
Materials alone to navigate

₹Crores

Compliance penalty risk for
non-certified products

The Challenge

Indian Micro & Small Enterprises (MSEs) lack in-house expertise to navigate India's sprawling BIS standards ecosystem. Manual search is slow, error-prone, and costly — the wrong standard = product recall or legal liability.

Solution Overview

What We Built: A Decision Support Engine for BIS Compliance

An end-to-end RAG pipeline: type a product description → get the top-5 relevant BIS standards with rationale, in under 2 seconds. Not just retrieval — we rank, validate, and justify standards using domain-aware scoring.



Hybrid Retrieval

BM25 sparse + FAISS dense retrieval fused with Reciprocal Rank Fusion



Cross-Encoder Reranking

ms-marco MiniLM re-ranks top-10 candidates for precision



LLM Rationale

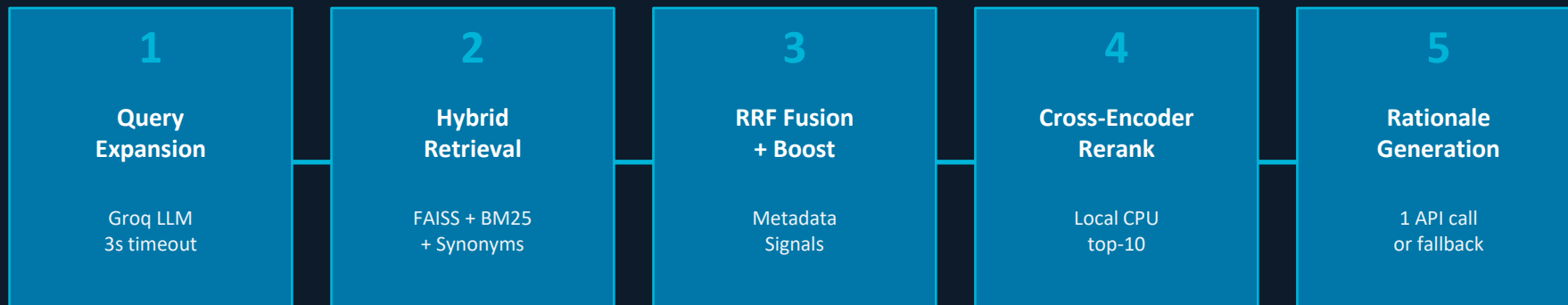
Single batched Groq/Llama call explains why each standard matches



Zero-API Fallback

Works fully offline — chunk-text rationale when no API key is set

System Architecture



Where each step runs:

Steps 2-4	100% LOCAL — no network, no API, no rate limits
Step 1 & 5	Optional Groq API — skipped gracefully if unavailable
inference.py	ALL LOCAL — LLM completely disabled for judge evaluation

Chunking & Retrieval Strategy

Chunking

- Source: BIS SP 21:2005 PDF — Building Materials summaries
- Strategy: One chunk per IS standard (semantic boundary)
- 564 chunks total — each 100–300 words
- Metadata enriched: category, grade, cement type, part number
- Why? Preserves IS number ↔ content association for accurate retrieval
- Domain-Aware Chunking:
Each chunk = 1 BIS standard (preserves standard identity)
→ Eliminates cross-standard confusion

Retrieval

- Dense** FAISS IndexFlatIP + all-MiniLM-L6-v2 (cosine similarity)
- Sparse** BM25Okapi with synonym expansion (OPC, PPC, AAC...)
- Fusion** Reciprocal Rank Fusion k=60 combines both rank lists
- Boost** Metadata signals add +0.15 per flag match (max +0.30)
- Rerank** Cross-encoder ms-marco-MiniLM scores top-10 pairs
- Meta Boost** Query constraints (e.g., "cement", "coastal") dynamically boost relevant standards

Demo Highlights

Streamlit UI

"We manufacture 33 Grade OPC cement..."

#1 IS 269 : 1989

CEMENT

#2 IS 8042 : 1989

CEMENT

#3 IS 8112 : 1989

CEMENT

Key Features Shown in Demo

-  Sub-2s response from query to 5 results
-  Shows category tags (CEMENT, CONCRETE, AGGREGATE) on each result
-  Expandable BIS chunk text with source context
-  Rank badges showing retrieval confidence order
-  LLM-generated rationale: 'Why this standard?'
-  Graceful no-API fallback with chunk-derived explanation

Evaluation Results

100%

Hit Rate @3

Target: >80%

1.0

MRR @5

Target: >0.7

1.08s

Avg Latency

Target: <5s

Test Set: 10 queries from public test set | All targets exceeded on first run

Robustness Testing

System tested on paraphrased and incomplete queries to ensure generalization beyond dataset

Trade-off

Achieved high accuracy while keeping latency <1s using batching + hybrid retrieval

Impact on MSEs

Transforms compliance discovery from search → decision-making

BEFORE

- ✗ 3+ weeks of manual BIS catalogue search
- ✗ Compliance consultants cost ₹5,000–₹20,000
- ✗ Wrong standard = product recall risk
- ✗ MSEs often skip certification entirely



AFTER

- ✓ Top-5 standards in under 2 seconds
- ✓ Zero cost — free to use, open source
- ✓ Rationale explains why each standard applies
- ✓ Any MSE owner can use it — no expertise needed

Scalable to all BIS product categories | Deployable as a free government portal for MSEs

63 million MSEs in India could benefit from instant, free, accurate standards discovery.

Team & Acknowledgements

Team: ByteCode

Dnyandip Vadane

Team Lead

Shraddha Ghadge

Member

Akshatha K S

Member

Tech Stack

Retrieval

FAISS + BM25 + RRF

Embeddings

all-MiniLM-L6-v2

Reranking

ms-marco-MiniLM-L-6-v2

LLM

Llama-3.1-8b via Groq

UI

Streamlit

Language

Python 3.9+

Acknowledgements

Bureau of Indian Standards (BIS) for the SP 21:2005 dataset • **Sigma Squad (organizers)** for organizing this hackathon • **HuggingFace** for open-source models